

Semester 2 Examinations 2019/2020

In submitting this work, I confirm that it is entirely my own. I acknowledge that I may be invited to an online interview if there is any concern in relation to the integrity of my exam, and I am aware that any breach will be subject to the University's Procedures for dealing with breaches of Exam Regulations:
<https://www.nuigalway.ie/media/registry/exams/QA230---Procedures-for-Dealing-with-Breaches-of-Examination-Regulations.pdf>

This Examination, MP491, is an Open Book examination.

Exam Code(s) 3BMS2, 3BS9, 4BME1, 4BPT2, 4FM2, 4BS2
Exam(s) 3rd & 4th Science, 4th Financial Mathematics and Economics, 4th Mathematics and Education, 4th Arts

Module Code(s) MP491
Module(s) Nonlinear Systems

Paper No. 1
Repeat Paper No

External Examiner(s) Professor Y. Fu
Internal Examiner(s) Dr P. Piironen, Dr M. Meere

Instructions: Marks are awarded for Question 1 in Section A and the TWO best questions in Section B.
All questions carry the same weight.

Duration 2 hours
No. of Pages 4
Discipline(s) Applied Mathematics
Course Co-ordinator(s) Dr Petri Piironen

Requirements:
Release in Exam Venue Yes
Statistical/ Log Tables Yes
Other Materials Calculator

Section A - Answer all parts (a)-(d). (5 Marks/part)

1. (a) Consider the nonlinear ordinary differential equation

$$\frac{dx}{dt} = -2x^2 + 8x - 6.$$

Find all equilibrium points and determine their stability.

- (b) Consider the linear system of ordinary differential equations

$$\begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} = \begin{pmatrix} 2 & -1 \\ 3 & -2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}.$$

Classify the equilibrium point $(x, y) = (0, 0)$ and sketch the phase-plane portrait.

- (c) Consider the system of nonlinear ordinary differential equations given by

$$\begin{aligned} \frac{dx}{dt} &= x^2 + 3y^2 - 2xy - 1, \\ \frac{dy}{dt} &= y^2 + 3x^2 - 2xy. \end{aligned}$$

- Show that the system is Hamiltonian.
- Find the Hamiltonian function.

- (d) Consider the map

$$x_{n+1} = -(x_n)^2 + 5x_n - 3, \quad n = 0, 1, 2, \dots$$

Find all fixed points and determine their linear stability.

Section B - Answer any two questions.

2. (a) Consider the one-parameter family of nonlinear ordinary differential equations

$$\frac{dx}{dt} = (x-1)(x-a^2+1),$$

where a is a real parameter.

- i. Find all equilibrium points.
- ii. Sketch the bifurcation diagram, and indicate the behaviour of non-equilibrium solutions using appropriate arrows.
- iii. Find all bifurcation points and classify them.

10 Marks

- (b) Consider the second order differential equation

$$\ddot{z} - (\mu - z^2)\dot{z} + z = 0. \quad (1)$$

- i. Show that (1) can be written as the system of ordinary differential equations

$$\begin{aligned} \dot{x} &= y, \\ \dot{y} &= -x + (\mu - x^2)y. \end{aligned}$$

- ii. Find the values of $\mu \in \mathbb{R}$ for which the origin $(x, y) = (0, 0)$ is a spiral.
- iii. Find the value of $\mu \in \mathbb{R}$ for which the origin $(x, y) = (0, 0)$ undergoes a Hopf bifurcation.

10 Marks

3. Consider the system of ordinary differential equations

$$\begin{aligned} \frac{dx}{dt} &= y - 1, \\ \frac{dy}{dt} &= x^2 + y - 2. \end{aligned}$$

- (a) Locate and classify all equilibrium points in the associated phase plane.

4 Marks



3

3. Consider the system of ordinary differential equations

$$\begin{aligned}\frac{dx}{dt} &= y - 1, \\ \frac{dy}{dt} &= x^2 + y - 2.\end{aligned}$$

(a) Locate and classify all equilibrium points in the associated phase plane.

4 Marks

(b) Identify, and sketch separately, the isoclines corresponding to $k = -1, 0, 1, \infty$, where

$$\frac{dy}{dx} = k.$$

12 Marks

(c) Use all four isoclines found in (b) to illustrate the phase-plane portrait for this system.

4 Marks

4. Consider the one-dimensional map

$$x_{n+1} = \frac{ax_n}{1 + x_n},$$

where $n = 0, 1, 2, \dots$ and a is a real parameter.

(a) Find all fixed points and investigate for what values of a the map only has non-negative fixed points.

5 Marks

(b) Calculate the stability of the fixed points found in (a) for $-\infty < a < \infty$.

5 Marks

(c) Sketch the bifurcation diagram for $-3 \leq a \leq 3$, where you only need to include the fixed points.

5 Marks

(d) Use the results in (a)-(c) and determine if x can undergo any bifurcations when $x > 0$. Explain your answer.

5 Marks

4

